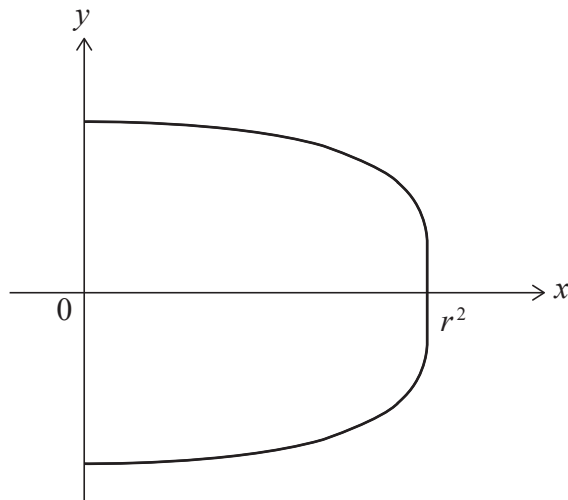


7. [Maximum mark: 6]

The curve $x^2 + y^4 = r^4$, where $0 \leq x \leq r^2$, is shown in the following diagram.



The region enclosed by the curve and the y -axis is rotated through 2π radians about the y -axis to form a solid of revolution.

Find an expression for the volume of the solid in the form $V = a\pi r^b$, where $a, b \in \mathbb{Q}^+$.

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